Elementary Analysis

Chapter 2: Sections 11–12

All 25 exercises, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Section 11: Exercises 11.1–11.11. Section 12: Exercises 12.1–12.14.

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11 Subsequences

A subsequence has the form $(s_{n_k})_{k\in\mathbb{N}}$, where its original indices satisfy $n_1 < n_2 < \cdots$. The selection map is $\sigma(k) = n_k$. The index k counts selected terms, whereas n_k is their position in the original sequence.

Subsequential limits in this section may be real numbers or $+\infty$ or $-\infty$. A proposed list must account for every possible subsequential limit. If S denotes the set of those limits, we use

$$\limsup s_n = \sup S, \qquad \liminf s_n = \inf S,$$

as established in Theorem 11.8. Increasing and decreasing remain weak notions, so a constant subsequence is monotone.

11.1 Exercise 11.1: A constant subsequence of a two-valued sequence

What we must show. For $a_n = 3+2(-1)^n$, list the first eight terms and give a constant subsequence together with its selection function.

Plan. Separate even and odd indices. The selection function must choose positive integer indices in strictly increasing order; specifying a repeated value without its indices is not enough.

(a) Compute the first eight terms.

At an odd index, $(-1)^n = -1$, so $a_n = 3 + 2(-1) = 3 - 2 = 1$. At an even index, $(-1)^n = 1$, so $a_n = 3 + 2(1) = 3 + 2 = 5$. Substituting the first eight indices gives

n	1	2	3	4	5	6	7	8
$(-1)^n$	-1	1	-1	1	-1	1	-1	1
a_n	1	5	1	5	1	5	1	5

(b) Specify and verify a selection function.

Define $\sigma : \mathbb{N} \to \mathbb{N}$ by $\sigma(k) = 2k$. For every $k \geq 1$, $\sigma(k) \geq 2 > 0$, and

$$\sigma(k+1) - \sigma(k) = 2(k+1) - 2k = 2 > 0.$$

Thus this is a valid strictly increasing sequence of indices. The selected values are

$$a_{\sigma(k)} = a_{2k} = 3 + 2(-1)^{2k} = 3 + 2 = 5 \qquad (k \in \mathbb{N}).$$

They form the constant subsequence $(5, 5, 5, \ldots)$.

Conclusion. The first eight terms are $1, 5, 1, 5, 1, 5, 1, 5$. One constant subsequence is $a_{2k} = 5$, selected by $\sigma(k) = 2k$. Selecting $\sigma(k) = 2k - 1$ would instead give the constant value 1.

11.2 Exercise 11.2: Five questions for each of four elementary sequences

What we must show. For each of

$$a_n = (-1)^n, \qquad b_n = \frac{1}{n}, \qquad c_n = n^2, \qquad d_n = \frac{6n+4}{7n-3},$$

answer all five questions: (a) a monotone subsequence with its indices; (b) the entire set of subsequential limits; (c) the limsup and liminf; (d) finite convergence or divergence to either infinity; and (e) boundedness. Subsequential limits here include the symbols $+\infty$ and $-\infty$ as in Definition 11.6.

Plan. We first record the tools used throughout Exercises 11.2–11.4. Each sequence then receives its own answers (a)–(e). In particular, constructing some subsequential limits will always be followed by a reason that no additional ones have been missed.

Tools shared by the three catalogues

Tool 1: Selection indices tend to infinity.

For any strictly increasing positive integer indices $n_1 < n_2 < \cdots$, one has $n_k \geq k$. At $k = 1$, $n_1 \geq 1$. If $n_k \geq k$, then the integer $n_{k+1} > n_k$ satisfies $n_{k+1} \geq n_k + 1 \geq k + 1$. Induction proves the claim. Consequently $k > N$ implies $n_k > N$.

The identity selection $n_k = k$ is allowed: a subsequence may retain all the terms of the original sequence. A constant subsequence is monotone under the book's non-strict definition, since its consecutive values satisfy both $u_{k+1} \geq u_k$ and $u_{k+1} \leq u_k$.

Tool 2: Every subsequence inherits an existing finite or infinite limit.

For a finite limit, this is Theorem 11.3. Its proof uses Tool 1: once every original index $n > N$ satisfies an error bound, every selected index n_k with $k > N$ satisfies it too. The same index argument proves the infinite cases directly. If $u_n \to +\infty$ and $M > 0$, choose N with $n > N \Longrightarrow u_n > M$. For $k > N$, $n_k \geq k > N$ gives $u_{n_k} > M$, so $u_{n_k} \to +\infty$. If $u_n \to -\infty$ and $M < 0$, choose N with $n > N \Longrightarrow u_n < M$; the identical implication $n_k \geq k > N$ gives $u_{n_k} < M$ and hence the negative infinite limit.

These alternatives are mutually exclusive. A finite limit L puts all sufficiently late terms in $(L-1, L+1)$, which prevents either infinite limit. The two signs of infinity are incompatible because their threshold tests with 1 and -1 would require a late term to be both greater than 1 and less than -1. Finite limits are unique by Theorem 7.3. Thus, if the original sequence has any existing limit L, its set of subsequential limits is exactly $\{L\}$: every subsequence has that limit, and the identity selection itself supplies a subsequence.

Tool 3: A sequence with only finitely many values.

Suppose every term belongs to a finite set $F = \{f_1, \ldots, f_m\}$ and every member of F occurs infinitely often. For each particular sequence below, we will display constant selections proving that every member of F is a subsequential limit. There can be no other finite subsequential limit. Indeed, if $L \notin F$, then the finite set of positive distances $\{|L - f_i| : 1 \leq i \leq m\}$ has a positive minimum δ. Every term, including every selected term, satisfies $|u_n - L| \geq \delta$, so convergence to L fails the error test with $\varepsilon = \delta/2$.

There are no infinite subsequential limits either. With $C = \max\{|f_1|, \ldots, |f_m|\}$, every term lies between $-C$ and C. Neither the positive threshold $C + 1$ nor the negative threshold $-C - 1$ can ever be crossed in the required direction. Therefore the full subsequential-limit set is precisely F.

Tool 4: Extract limsup, liminf, and full-sequence behavior from the set.

If S is the full set of subsequential limits, Theorem 11.8(ii) gives

$$\limsup_{n\to\infty} u_n = \sup S, \qquad \liminf_{n\to\infty} u_n = \inf S.$$

In the explicit sets below these are their greatest and least elements in the extended order. Theorem 11.8(iii) says that the original sequence has a limit if and only if S has exactly one element. A singleton real element means finite convergence; a singleton $\{+\infty\}$ or $\{-\infty\}$ means divergence of the corresponding sign. If S contains two or more elements, there is no finite limit and neither infinite limit exists.

The sequence $a_n = (-1)^n$

(a) A monotone subsequence. Choose $n_k = 2k$. The indices start at 2 and increase by 2, so they are positive and strictly increasing. Then $a_{n_k} = (-1)^{2k} = 1$ at every k. This constant subsequence is monotone.

(b) All subsequential limits. At the indices $2k-1$ the values are -1, and at the indices $2k$ they are 1. Both selections have positive indices and successive differences 2. Thus constant subsequences give both -1 and 1 as limits. The original sequence takes only these two values, so Tool 3 excludes every other finite or infinite subsequential limit:

$$S_a = \{-1, 1\}.$$

(c) Limsup and liminf. The greatest element of S_a is 1 and the least is -1. Tool 4 therefore gives $\limsup a_n = 1$ and $\liminf a_n = -1$.

(d) Full-sequence behavior. Since S_a has two distinct elements, Tool 4 excludes all three possibilities: (a_n) does not converge to a real number, does not diverge to $+\infty$, and does not diverge to $-\infty$.

(e) Boundedness. Every term is either -1 or 1, so $-1 \le a_n \le 1$ for every n. Thus (a_n) is bounded.

The sequence $b_n = 1/n$

(a) A monotone subsequence. Take the identity selection $n_k = k$, whose indices start at 1 and increase by 1. It gives $b_{n_k} = 1/k$. Its consecutive differences are

$$\frac{1}{k+1} - \frac{1}{k} = \frac{k-(k+1)}{k(k+1)} = -\frac{1}{k(k+1)} < 0.$$

The denominator is positive for $k \ge 1$, so the selected sequence is strictly decreasing.

(b) All subsequential limits. The original sequence converges to 0: given $\varepsilon > 0$, $n > 1/\varepsilon$ implies $|1/n - 0| = 1/n < \varepsilon$. Tool 2 therefore proves that every subsequence has limit 0, and excludes any additional finite or infinite limit. Thus $S_b = \{0\}$.

(c) Limsup and liminf. The supremum and infimum of $\{0\}$ are both 0. Hence $\limsup b_n = \liminf b_n = 0$ by Tool 4.

(d) Full-sequence behavior. The preceding error estimate proves finite convergence to 0. It does not diverge to either sign of infinity, by the mutual exclusivity recorded in Tool 2.

(e) Boundedness. For every positive integer n, $0 < 1/n \le 1$. Thus $0 < b_n \le 1$, providing a lower and an upper bound.

The sequence $c_n = n^2$

(a) A monotone subsequence. Take $n_k = k$. Then $c_{n_k} = k^2$, and

$$(k+1)^2 - k^2 = k^2 + 2k + 1 - k^2 = 2k + 1 > 0.$$

Thus the identity subsequence is strictly increasing.

(b) All subsequential limits. The original sequence has limit $+\infty$: for $M > 0$, $n > \sqrt{M}$ implies $n^2 > M$. Tool 2 proves that every subsequence has this same infinite limit. Consequently the exact set is $S_c = \{+\infty\}$; in particular, no finite limit or negative-infinity limit has been omitted.

(c) Limsup and liminf. Both the greatest and least elements of this singleton are $+\infty$. Thus $\limsup c_n = \liminf c_n = +\infty$.

(d) Full-sequence behavior. The sequence diverges to $+\infty$. It has no finite real limit and does not diverge to $-\infty$.

(e) Boundedness. It is bounded below by 1, since $n \geq 1$. It is not bounded above: for any proposed upper bound B, choose $n > \max\{1, B\}$. Then $n^2 \geq n > B$. Thus (c_n) is not a bounded sequence.

The sequence $d_n = (6n + 4)/(7n - 3)$

For every $n \geq 1$, $7n - 3 \geq 4 > 0$, so each term is defined with a positive denominator. An exact subtraction identifies both its limit and its side of approach:

$$\begin{aligned} d_n - \frac{6}{7} &= \frac{7(6n+4) - 6(7n-3)}{7(7n-3)} \\ &= \frac{42n + 28 - 42n + 18}{7(7n-3)} \\ &= \frac{46}{7(7n-3)} > 0. \end{aligned}$$

(a) A monotone subsequence. Use $n_k = k$. Subtract consecutive values using the error formula just established:

$$\begin{aligned} d_{k+1} - d_k &= \frac{46}{7}\left(\frac{1}{7k+4} - \frac{1}{7k-3}\right) \\ &= \frac{46}{7}\,\frac{(7k-3) - (7k+4)}{(7k+4)(7k-3)} \\ &= \frac{46}{7}\,\frac{-7}{(7k+4)(7k-3)} \\ &= -\frac{46}{(7k+4)(7k-3)} < 0. \end{aligned}$$

Both denominator factors are positive for $k \geq 1$. Thus the full sequence, and hence the identity subsequence, is strictly decreasing.

(b) All subsequential limits. For $n \geq 1$, $7n - 3 \geq 4n$ because $(7n - 3) - 4n = 3(n - 1) \geq 0$. Therefore

$$\left|d_n - \frac{6}{7}\right| = \frac{46}{7(7n-3)} \leq \frac{46}{28n} = \frac{23}{14n}.$$

For $\varepsilon > 0$, any threshold $N > 23/(14\varepsilon)$ makes this error less than ε for $n > N$. Thus $d_n \to 6/7$. Tool 2 supplies exhaustiveness, so $S_d = \{6/7\}$.

(c) Limsup and liminf. Tool 4 applied to the singleton gives $\limsup d_n = \liminf d_n = 6/7$.

(d) Full-sequence behavior. The sequence converges to the finite value $6/7$. Neither sign of infinite divergence is possible.

(e) Boundedness. The positive error formula shows $d_n > 6/7$ for every n. Strict decrease shows no term is larger than the first:

$$d_1 = \frac{6+4}{7-3} = \frac{10}{4} = \frac{5}{2}.$$

Thus $6/7 < d_n \leq 5/2$ for every n, so the sequence is bounded.

Conclusion. In Exercise 11.2, (b_n) and (d_n) converge, to 0 and $6/7$ respectively; (c_n) diverges to $+\infty$; and (a_n) has no extended limit. None diverges to $-\infty$. Exactly $(a_n), (b_n), (d_n)$ are bounded. Each of the four sequences has been given a monotone selection and its complete subsequential-limit set above.

11.3 Exercise 11.3: Five questions for four bounded sequences

What we must show. Answer (a)–(e) of Exercise 11.2 separately for

$$s_n = \cos(n\pi/3), \qquad t_n = \frac{3}{4n+1}, \qquad u_n = \left(-\frac{1}{2}\right)^n, \qquad v_n = (-1)^n + \frac{1}{n}.$$

We continue to use the four shared tools, so all sets of subsequential limits are understood in the extended sense.

The sequence $s_n = \cos(n\pi/3)$

The cosine values are determined by the residue of n modulo 6. Write $n = 6q + r$, with $q \geq 0$ an integer and $r \in \{1, 2, 3, 4, 5, 6\}$. Then $n\pi/3 = 2q\pi + r\pi/3$, so the period 2π gives

r	1	2	3	4	5	6
$\cos(r\pi/3)$	$1/2$	$-1/2$	-1	$-1/2$	$1/2$	1

(a) A monotone subsequence. Select $n_k = 6k$. These positive indices start at 6 and increase by 6. The selected terms satisfy $s_{6k} = \cos(2k\pi) = 1$, so this is a constant monotone subsequence.

(b) All subsequential limits. The table gives precisely four different values. The following explicit selections show that each occurs as a constant subsequence:

indices n_k	first index	s_{n_k}
$6k-5$	1	$1/2$
$6k-4$	2	$-1/2$
$6k-3$	3	-1
$6k$	6	1

Each first index is positive and each sequence of indices increases by 6. The period table proves there are no other values; Tool 3 excludes any other finite or infinite subsequential limit. Hence

$$S_s = \{-1, -1/2, 1/2, 1\}.$$

(c) Limsup and liminf. The greatest element is 1 and the least is -1, so $\limsup s_n = 1$ and $\liminf s_n = -1$.

(d) Full-sequence behavior. The set S_s has four distinct elements, rather than one. Tool 4 shows that the original sequence does not converge to a finite value and does not diverge to either infinity.

(e) Boundedness. The table puts every value between -1 and 1, so $-1 \le s_n \le 1$ for every n and the sequence is bounded.

The sequence $t_n = 3/(4n+1)$

(a) A monotone subsequence. The denominators are positive because $4n+1 \ge 5$. Choose $n_k = k$. The difference between consecutive selected terms is

$$\begin{aligned} t_{k+1} - t_k &= \frac{3}{4k+5} - \frac{3}{4k+1} \\ &= \frac{3[(4k+1)-(4k+5)]}{(4k+5)(4k+1)} \\ &= -\frac{12}{(4k+5)(4k+1)} < 0. \end{aligned}$$

Thus the identity subsequence is strictly decreasing.

(b) All subsequential limits. For every $n \ge 1$,

$$0 < t_n = \frac{3}{4n+1} \le \frac{3}{4n}.$$

For $\varepsilon > 0$, $n > 3/(4\varepsilon)$ makes the last expression less than ε. Hence $t_n \to 0$. Tool 2 proves that all subsequences have the same limit, so $S_t = \{0\}$, with no additional extended limits.

(c) Limsup and liminf. Both extremal elements of this singleton are 0. Thus $\limsup t_n = \liminf t_n = 0$.

(d) Full-sequence behavior. The sequence converges to 0 and diverges to neither $+\infty$ nor $-\infty$.

(e) Boundedness. The inequality $4n+1 \ge 5$ gives $0 < t_n \le 3/5$. Therefore the sequence is bounded.

The sequence $u_n = (-1/2)^n$

(a) A monotone subsequence. Choose the even indices $n_k = 2k$, positive and increasing by 2. Then

$$u_{2k} = \left(-\frac{1}{2}\right)^{2k} = \left(\frac{1}{4}\right)^k > 0.$$

The next selected term is $u_{2(k+1)} = (1/4)u_{2k}$. Since $0 < 1/4 < 1$ and $u_{2k} > 0$, it is strictly smaller than u_{2k}. Thus this subsequence is strictly decreasing.

(b) All subsequential limits. The base has magnitude $1/2 < 1$. Theorem 9.7(b), or Exercise 9.13, gives $(-1/2)^n \to 0$. Equivalently its distance from 0 is $(1/2)^n \to 0$, independently of the alternating sign. Tool 2 then gives the exhaustive set $S_u = \{0\}$.

(c) Limsup and liminf. The singleton has both supremum and infimum 0, so $\limsup u_n = \liminf u_n = 0$.

(d) Full-sequence behavior. It converges to the finite value 0, and diverges to neither sign of infinity.

(e) Boundedness. For $n \geq 1$,

$$|u_n| = \left(\frac{1}{2}\right)^n = \frac{1}{2}\left(\frac{1}{2}\right)^{n-1} \leq \frac{1}{2}.$$

The final product is at most $1/2$ because its remaining $n-1$ factors are at most 1; at $n=1$ that remaining product is empty and equals 1. Hence $-1/2 \leq u_n \leq 1/2$, a bound valid at every index.

The sequence $v_n = (-1)^n + 1/n$

(a) A monotone subsequence. Choose $n_k = 2k$, again positive and increasing by 2. The selected terms are $v_{2k} = 1 + 1/(2k)$, and

$$\begin{aligned} v_{2(k+1)} - v_{2k} &= \frac{1}{2k+2} - \frac{1}{2k} \\ &= \frac{2k - (2k+2)}{(2k+2)(2k)} \\ &= -\frac{1}{2k(k+1)} < 0. \end{aligned}$$

This is a strictly decreasing subsequence.

(b) All subsequential limits. The even and odd selections respectively give

$$v_{2k} = 1 + \frac{1}{2k} \longrightarrow 1, \qquad v_{2k-1} = -1 + \frac{1}{2k-1} \longrightarrow -1.$$

Both index formulas start at positive integers and increase by 2. The reciprocal errors tend to 0: for example, $1/(2k) \leq 1/k$ and $1/(2k-1) \leq 1/k$ for $k \geq 1$. Thus -1 and 1 are both subsequential limits.

To exclude every other finite candidate, let $L \notin \{-1, 1\}$ and set

$$\delta = \min\{|L-1|, |L+1|\} > 0.$$

Consider any subsequence with indices n_k. Since $n_k \geq k$, we have $1/n_k \leq 1/k$. For $k > 2/\delta$, this is less than $\delta/2$. The reverse triangle inequality then yields

$$\begin{aligned} |v_{n_k} - L| &= |((-1)^{n_k} - L) + 1/n_k| \\ &\geq |(-1)^{n_k} - L| - \frac{1}{n_k} \\ &\geq \delta - \frac{1}{n_k} > \frac{\delta}{2}. \end{aligned}$$

This defeats the convergence test with error tolerance $\delta/2$ for every such subsequence. Finally,

$$|v_n| \leq |(-1)^n| + \frac{1}{n} = 1 + \frac{1}{n} \leq 2,$$

so no subsequence can tend to either infinity. The complete set is therefore $S_v = \{-1, 1\}$.

(c) Limsup and liminf. Tool 4 gives $\limsup v_n = 1$ and $\liminf v_n = -1$. The values of the sequence approach these numbers along the even and odd selections, respectively.

(d) Full-sequence behavior. There are two distinct subsequential limits. Thus the full sequence has no finite limit and diverges to neither infinity.

(e) Boundedness. The absolute-value estimate $|v_n| \le 2$ already proves boundedness. A sharper bound follows directly from parity: at even indices $n = 2k$, $1 < v_{2k} = 1 + 1/(2k) \le 3/2$; at odd indices $n = 2k-1$, $-1 < v_{2k-1} = -1 + 1/(2k-1) \le 0$. Combining the cases gives $-1 < v_n \le 3/2$ for every n.

Conclusion. In Exercise 11.3 all four sequences are bounded. The sequences (t_n) and (u_n) converge to 0; (s_n) and (v_n) have no extended limit. None diverges to either infinity. Their exact subsequential-limit sets are, in order, $\{-1, -1/2, 1/2, 1\}$, $\{0\}$, $\{0\}$, and $\{-1, 1\}$, with limsup and liminf the largest and smallest elements of these sets.

11.4 Exercise 11.4: Five questions for oscillation with and without growth

What we must show. Answer (a)–(e) of Exercise 11.2 separately for

$$w_n = (-2)^n, \qquad x_n = 5^{(-1)^n}, \qquad y_n = 1 + (-1)^n, \qquad z_n = n\cos(n\pi/4).$$

In the second expression, $(-1)^n$ is the *exponent* of 5; the expression is not multiplication by $(-1)^n$.

Plan. Parity resolves the first three sequences. For the last, a complete eight-residue table identifies the zero terms and the growing positive and negative terms. A separate bounded-tail argument then excludes any unlisted finite subsequential limit.

The sequence $w_n = (-2)^n$

(a) A monotone subsequence. Take $n_k = 2k$, whose positive indices increase by 2. Then $w_{2k} = 4^k > 0$. The next selected value is $4^{k+1} = 4 \cdot 4^k > 4^k$, so this subsequence is strictly increasing.

(b) All subsequential limits. The even selection gives $w_{2k} = 4^k \to +\infty$ by Exercise 9.13, since the base 4 is greater than 1. The odd selection $n_k = 2k - 1$ gives

$$w_{2k-1} = -2^{2k-1} = -\frac{1}{2}\,4^k \longrightarrow -\infty$$

by the negative scaling result of Exercise 9.10. Both selections are valid positive increasing indices.

There is no finite subsequential limit. Indeed, $|w_n| = 2^n \to +\infty$ by Exercise 9.13. Tool 2 applied to this positive magnitude sequence gives $|w_{n_k}| \to +\infty$ for every selection. If $w_{n_k} \to L \in \mathbb{R}$, the error tolerance 1 would give, eventually,

$$|w_{n_k}| \le |w_{n_k} - L| + |L| < 1 + |L|\,,$$

contradicting that magnitude limit. The only remaining extended candidates are the two already constructed, so

$$S_w = \{-\infty, +\infty\}.$$

(c) Limsup and liminf. The extremal elements of S_w give $\limsup w_n = +\infty$ and $\liminf w_n = -\infty$.

(d) Full-sequence behavior. Since S_w has two elements, the full sequence has no extended limit. In particular it does not converge to a real number and does not diverge to either single sign of infinity. Its large positive and negative terms do not constitute a limit of one sign.

(e) Boundedness. The even terms exceed every real upper bound and the odd terms go below every real lower bound, by their infinite limits just proved. Hence (w_n) is bounded neither above nor below, and is not a bounded sequence.

The sequence $x_n = 5^{(-1)^n}$

At even indices the exponent equals 1, and at odd indices it equals -1. Therefore

$$x_{2k} = 5^1 = 5, \qquad x_{2k-1} = 5^{-1} = \frac{1}{5}.$$

(a) A monotone subsequence. Choose $n_k = 2k$, positive and increasing by 2. Then $x_{n_k} = 5$ at every index, a constant monotone subsequence.

(b) All subsequential limits. The even and odd index selections are constant with values 5 and $1/5$ respectively, so both are subsequential limits. They are the only values the original sequence ever takes; Tool 3 therefore excludes all other finite and infinite subsequential limits. Thus $S_x = \{1/5, 5\}$.

(c) Limsup and liminf. Since $1/5 < 5$, Tool 4 gives $\limsup x_n = 5$ and $\liminf x_n = 1/5$.

(d) Full-sequence behavior. The two distinct subsequential limits show that (x_n) does not converge to a finite number and diverges to neither sign of infinity.

(e) Boundedness. Every term is either $1/5$ or 5, so $1/5 \le x_n \le 5$. Both bounding values are attained, and the sequence is bounded.

The sequence $y_n = 1 + (-1)^n$

(a) A monotone subsequence. Select $n_k = 2k$, positive and increasing by 2. Then $y_{2k} = 1 + 1 = 2$ at every selected index. This constant subsequence is monotone.

(b) All subsequential limits. At odd indices, $y_{2k-1} = 1 - 1 = 0$, while at even indices $y_{2k} = 2$. The two valid selections give constant subsequences with limits 0 and 2. Because these are the only values taken by any term, Tool 3 proves the exhaustive set $S_y = \{0, 2\}$, with no infinite subsequential limit.

(c) Limsup and liminf. The largest and smallest elements give $\limsup y_n = 2$ and $\liminf y_n = 0$.

(d) Full-sequence behavior. The set of subsequential limits has two distinct elements. Thus there is no finite limit and no divergence to either $+\infty$ or $-\infty$.

(e) Boundedness. The values satisfy $0 \le y_n \le 2$ at every index, so the sequence is bounded.

The sequence $z_n = n\cos(n\pi/4)$

Write $n = 8q + r$, with $q \ge 0$ an integer and $r \in \{1, 2, 3, 4, 5, 6, 7, 8\}$. Then $n\pi/4 = 2q\pi + r\pi/4$. The cosine period and its unit-circle values give the complete table

r	$\cos(r\pi/4)$	z_n when $n = 8q + r$
1	$1/\sqrt{2}$	$n/\sqrt{2}$
2	0	0
3	$-1/\sqrt{2}$	$-n/\sqrt{2}$
4	-1	$-n$
5	$-1/\sqrt{2}$	$-n/\sqrt{2}$
6	0	0
7	$1/\sqrt{2}$	$n/\sqrt{2}$
8	1	n

Thus every nonzero term has magnitude at least $n/\sqrt{2}$; the only other possibility is the exact value 0.

(a) A monotone subsequence. Take $n_k = 8k$, with first index 8 and successive difference $8 > 0$. The selected values are

$$z_{8k} = 8k\cos(2k\pi) = 8k.$$

Their consecutive differences equal $8 > 0$, so this is a strictly increasing subsequence.

(b) Construct the three candidate limits. The same even-period selection gives $z_{8k} = 8k \to +\infty$: given $M > 0$, every $k > M/8$ makes $8k > M$. For negative infinity, choose $n_k = 8k - 4$, starting at 4 and increasing by 8. Then

$$z_{8k-4} = (8k-4)\cos((2k-1)\pi) = -(8k-4) = -8k + 4.$$

For $M < 0$, every $k > (4-M)/8$ satisfies $-8k + 4 < M$, so this subsequence tends to $-\infty$.

For the finite value 0, choose $n_k = 4k - 2$, starting at 2 and increasing by 4. The angle is $(4k-2)\pi/4 = k\pi - \pi/2$, whose cosine is 0. Thus $z_{4k-2} = 0$ for every k, giving a constant subsequence with limit 0.

(b) Exclude every other finite limit. Suppose a subsequence z_{n_k} converges to a finite $L \in \mathbb{R}$. Convergence with error tolerance 1 gives a threshold K_1 such that for $k > K_1$,

$$|z_{n_k}| \le |z_{n_k} - L| + |L| < 1 + |L|\,.$$

Choose $K_2 > \sqrt{2}(1 + |L|)$. For $k > K_2$, Tool 1 gives $n_k \ge k > \sqrt{2}(1 + |L|)$, so

$$\frac{n_k}{\sqrt{2}} > 1 + |L|\,.$$

At an index with $k > \max\{K_1, K_2\}$, a nonzero selected term would have, by the period table, $|z_{n_k}| \ge n_k/\sqrt{2} > 1 + |L|$, contradicting the bounded-tail estimate. Thus every sufficiently late selected term must be exactly 0.

Such a subsequence has limit 0 directly from the definition. It cannot have a different finite limit $L \neq 0$, because its late errors from L would all equal $|L|$, failing the tolerance $\varepsilon = |L|/2$. Consequently 0 is the only possible finite subsequential limit. The only infinite possibilities, $+\infty$ and $-\infty$, have already been constructed. The complete set is

$$S_z = \{-\infty, 0, +\infty\}.$$

(c) Limsup and liminf. The greatest and least elements of this set are $+\infty$ and $-\infty$. Hence $\limsup z_n = +\infty$ and $\liminf z_n = -\infty$.

(d) Full-sequence behavior. There are three distinct subsequential limits. By Tool 4 the original sequence does not converge to a finite real number and does not diverge to either infinity. In particular, the growing magnitudes in some residue classes do not remove the infinitely many exact zero terms.

(e) Boundedness. The values $z_{8k} = 8k$ are unbounded above, and the values $z_{8k-4} = -8k + 4$ are unbounded below. Thus (z_n) is bounded neither above nor below, and is not a bounded sequence.

Conclusion. In Exercise 11.4 none of the four full sequences has a finite or infinite limit. The sequences (x_n) and (y_n) are bounded, while (w_n) and (z_n) are unbounded above and below. Their complete subsequential-limit sets are, respectively,

$$\{-\infty, +\infty\}, \qquad \{1/5, 5\}, \qquad \{0, 2\}, \qquad \{-\infty, 0, +\infty\}.$$

The limsup and liminf for each sequence are the greatest and least elements of its displayed set, as proved in the individual answers to (c).

11.5 Exercise 11.5: Subsequential limits of an enumeration of a rational interval

What we must show. Let (q_n) enumerate all rational numbers in $(0, 1]$. Find (a) its set of subsequential limits and (b) its limit superior and limit inferior.

Plan. Every term lies between 0 and 1, so a subsequential limit must lie in $[0, 1]$. Conversely, every neighborhood of a point of $[0, 1]$ contains infinitely many distinct rational values in $(0, 1]$. Removing a finite prefix of the enumeration cannot remove all those values. This will allow a construction with strictly increasing indices.

1. Explain the role of enumeration and distinct values

The enumeration hypothesis says that each rational number in $(0, 1]$ occurs somewhere in the list. It does not require a particular rational number to occur infinitely often. For example, the value 1 may occur only once. The proof below uses infinitely many *different* rational values near a proposed limit, so it works both with and without repeated terms.

Every nonempty open interval (c, d) contains infinitely many rational numbers. To prove this from density, suppose its rational numbers formed a finite set. Density ensures the set is nonempty, so it would have a largest member $r < d$. Density applied to (r, d) would produce a rational number larger than r still belonging to (c, d), contradicting the choice of r. Thus there are infinitely many distinct rational numbers in the interval.

2. Part (a): Find infinitely many eligible values near every target

Fix $t \in [0, 1]$ and $\varepsilon > 0$. Consider

$$c = \max\{0, t - \varepsilon\}, \qquad d = \min\{1, t + \varepsilon\}.$$

We have $c < d$. At $t = 0$, this follows from $c = 0 < \min\{1, \varepsilon\} = d$. At $t = 1$, it follows from $c = \max\{0, 1 - \varepsilon\} < 1 = d$. For $0 < t < 1$, both $0 < t$ and $t - \varepsilon < t$ give $c < t$, while $t < 1$ and $t < t + \varepsilon$ give $t < d$.

Every rational number $r \in (c, d)$ satisfies

$$0 < r < 1, \qquad t - \varepsilon < r < t + \varepsilon.$$

It therefore belongs to the enumerated set and satisfies $|r - t| < \varepsilon$. By Step 1, there are infinitely many distinct such rational values. A finite prefix $q_1, \ldots, q_N$ contains only finitely many values, so it cannot contain them all. Hence for every finite index bound N there is an index $n > N$ such that

$$|q_n - t| < \varepsilon.$$

This reasoning includes the endpoints $t = 0$ and $t = 1$.

3. Construct a subsequence converging to the chosen target

Put $n_0 = 0$. At stage $k \geq 1$, apply Step 2 with $\varepsilon = 1/k$ and with the finite index bound n_{k-1}. Choose an index $n_k > n_{k-1}$ satisfying

$$|q_{n_k} - t| < \frac{1}{k}.$$

For definiteness, one may take the least eligible index, since a nonempty set of positive integers has a least element. The inequalities $n_1 < n_2 < \cdots$ ensure that (q_{n_k}) is a subsequence.

Let $\eta > 0$ and choose an integer $K > 1/\eta$ by the Archimedean property. If $k > K$, then

$$|q_{n_k} - t| < \frac{1}{k} < \eta.$$

Thus $q_{n_k} \to t$. Since $t \in [0, 1]$ was arbitrary, every point of that interval is a subsequential limit. This is also the infinite-neighborhood criterion of Theorem 11.2(i).

4. Exclude every other finite or infinite limit

Every subsequence still satisfies $0 < q_{n_k} \leq 1$. If it had a finite limit $t < 0$, use the positive tolerance $\eta = -t/2$. Eventually its terms would be less than $t + \eta = t/2 < 0$, a contradiction. If it had a finite limit $t > 1$, use $\eta = (t-1)/2$. Eventually its terms would be greater than $t - \eta = (t+1)/2 > 1$, again a contradiction. Hence every finite subsequential limit lies in $[0, 1]$.

No subsequence can tend to $+\infty$, because its terms never exceed 1 and so cannot eventually exceed the positive threshold 2. No subsequence can tend to $-\infty$, because its terms are positive and so cannot eventually be less than -1. Combining this exclusion with Step 3 proves that the exact set of subsequential limits is $[0, 1]$.

5. Part (b): Take the extreme subsequential limits

By Theorem 11.8(ii), the limit superior is the supremum of the set of subsequential limits, and the limit inferior is its infimum. The interval $[0, 1]$ has maximum 1 and minimum 0, so

$$\limsup_{n\to\infty} q_n = 1, \qquad \liminf_{n\to\infty} q_n = 0.$$

Conclusion. (a) The set of subsequential limits is $[0, 1]$. (b) The limit superior is 1 and the limit inferior is 0. No repeated occurrence of an individual rational value is assumed.

11.6 Exercise 11.6: A subsequence of a subsequence

What we must show. Prove that a subsequence of a subsequence of a given sequence is itself a subsequence of the original sequence.

Plan. Use Definition 11.1 in terms of strictly increasing selection functions. Compose the two selection functions and verify that the composition remains strictly increasing.

1. Write the two selections precisely

Let $s : \mathbb{N} \to \mathbb{R}$ represent the original sequence. Suppose t is a subsequence of s. There is a strictly increasing function $\sigma : \mathbb{N} \to \mathbb{N}$ with

$$t_k = s_{\sigma(k)} \qquad (k \in \mathbb{N}).$$

Now suppose u is a subsequence of t. There is a strictly increasing function $\tau : \mathbb{N} \to \mathbb{N}$ with

$$u_j = t_{\tau(j)} \qquad (j \in \mathbb{N}).$$

Here strictly increasing refers to the *indices* selected by the functions; the numerical sequence values themselves need not increase or be distinct.

2. Form and check the composite selection

Define $\rho = \sigma \circ \tau$; explicitly, $\rho(j) = \sigma(\tau(j))$. Because τ takes positive integer values and σ is defined on every positive integer, ρ is a function from $\mathbb{N}$ to $\mathbb{N}$. If $j < \ell$, strict increase of τ gives $\tau(j) < \tau(\ell)$. Strict increase of σ then gives

$$\rho(j) = \sigma(\tau(j)) < \sigma(\tau(\ell)) = \rho(\ell).$$

Thus ρ is strictly increasing.

3. Identify the selected values

Substituting the first selection into the second yields, for each j,

$$u_j = t_{\tau(j)} = s_{\sigma(\tau(j))} = s_{\rho(j)}.$$

The values u_j therefore select terms of s in the strictly increasing index order $\rho(1) < \rho(2) < \cdots$. Definition 11.1 says exactly that this makes u a subsequence of s.

Conclusion. The composite selection function $\sigma \circ \tau$ proves that every subsequence of a subsequence is a subsequence of the original sequence.

11.7 Exercise 11.7: A subsequence of an enumeration of all rational numbers

What we must show. If (r_n) enumerates $\mathbb{Q}$, construct a subsequence (r_{n_k}) tending to $+\infty$.

Plan. Every tail of the enumeration remains unbounded above. Prove that statement with explicit control of a finite prefix, then select successive terms that exceed their subsequence indices.

1. Show that arbitrarily large rational values occur after any prefix

Fix a positive integer N and a real threshold M. The finite prefix has a finite maximum

$$P = \max\{r_1, \ldots, r_N\}.$$

By the Archimedean property, choose a positive integer $p > \max\{M, P, 1\}$. Since $p \in \mathbb{Q}$, the enumeration contains p: there is an index j with $r_j = p$. The inequality $p > P$ prevents $j \leq N$, because every term of that prefix is at most P. Thus $j > N$ and $r_j > M$. This proves that the tail after any finite prefix is unbounded above. Before any terms are excluded, the same conclusion follows by choosing an integer $p > M$ in the enumeration.

2. Select terms recursively with increasing indices

Choose n_1 so that $r_{n_1} > 1$; this is possible by Step 1. Suppose $n_1 < \cdots < n_{k-1}$ have been chosen. Apply Step 1 to the prefix ending at n_{k-1} and to the real threshold

$$M_k = \max\{k, r_{n_{k-1}}\}.$$

It supplies an index $n_k > n_{k-1}$ with

$$r_{n_k} > M_k \geq k, \qquad r_{n_k} > r_{n_{k-1}}.$$

The resulting indices are strictly increasing, so they define a subsequence. The selected values are strictly increasing as well, although this latter property is not needed just to be a subsequence.

3. Check the definition of divergence to positive infinity

Let $M > 0$ be arbitrary. Choose an integer $K > M$. If $k > K$, the construction gives

$$r_{n_k} > k > K > M.$$

Thus the selected values eventually exceed every prescribed positive real threshold. By Definition 9.8, $r_{n_k} \to +\infty$.

Conclusion. Such a subsequence exists; the construction ensures both $n_1 < n_2 < \cdots$ and $r_{n_k} > k$, which proves its limit is $+\infty$.

11.8 Exercise 11.8: Reflection exchanges the limit inferior and limit superior

What we must show. For every real sequence (s_n), prove

$$\liminf_{n\to\infty} s_n = -\limsup_{n\to\infty}(-s_n)$$

using Definition 10.6 and the reflection identity of Exercise 5.4.

Plan. Reflect each tail before taking its extremum. If the original sequence is unbounded below, use the infinite-value conventions in Definition 10.6 directly. Otherwise all relevant tail bounds are real numbers, and their limits can be compared using finite convergence or the definition of an infinite limit.

1. Apply reflection to the nonempty tails

For each positive integer N, let

$$E_N = \{s_n : n > N\}, \qquad -E_N = \{-s_n : n > N\}.$$

Each set is nonempty, since it contains the corresponding term with index $N + 1$. Exercise 5.4 gives the extended equality

$$\inf E_N = -\sup(-E_N).$$

Write $u_N = \inf E_N$ and $v_N = \sup(-E_N)$. The equality says $u_N = -v_N$, or equivalently $v_N = -u_N$, with the negation conventions of Section 5.

2. Handle a sequence that is unbounded below

Suppose (s_n) is unbounded below. Every tail E_N is also unbounded below. Indeed, if some E_N had a real lower bound c, then

$$\min\{c, s_1, \ldots, s_N\}$$

would be a real lower bound for the entire sequence, a contradiction. Hence $u_N = -\infty$ for every N.

The reflected sequence $(-s_n)$ is unbounded above: for every real M, a term $s_n < -M$ yields $-s_n > M$. Thus $v_N = +\infty$ for every N. Definition 10.6 explicitly decrees, in these respective cases,

$$\liminf s_n = -\infty, \qquad \limsup(-s_n) = +\infty.$$

Consequently

$$-\limsup(-s_n) = -(+\infty) = -\infty = \liminf s_n.$$

This step uses the stated conventions, not a real-sequence limit law applied to a sequence of infinite symbols.

3. When the original sequence is bounded below, all these tail bounds are real

Suppose $c \in \mathbb{R}$ is a lower bound for all the s_n. Then every nonempty tail E_N is bounded below by c, so $u_N \in \mathbb{R}$ by Corollary 4.5. The reflected tail is bounded above by $-c$, so $v_N \in \mathbb{R}$ by completeness. The identity from Step 1 is now an identity of real numbers:

$$v_N = -u_N \qquad (N \in \mathbb{N}).$$

Moreover $E_{N+1} \subseteq E_N$, and restricting the set cannot decrease its infimum. Thus $u_N \leq u_{N+1}$. The real sequence (u_N) is increasing and is bounded below by c. The monotone-sequence results of Section 10 show that it either converges to a finite real number or diverges to $+\infty$. These are the only two cases to consider here.

4. Treat a finite limit of the real tail infima

If $u_N \to u \in \mathbb{R}$, then Theorem 9.2, applied to the real sequence (u_N) with scalar -1, gives $v_N = -u_N \to -u$. Definition 10.6 now identifies

$$\liminf s_n = u, \qquad \limsup(-s_n) = -u.$$

Negating the second real equality proves $-\limsup(-s_n) = u = \liminf s_n$.

5. Treat divergence of the real tail infima to positive infinity

If $u_N \to +\infty$, we show directly that $v_N = -u_N \to -\infty$. Let $M < 0$. The number $-M$ is positive, so the hypothesis supplies a threshold N_0 such that $N > N_0$ implies $u_N > -M$. Negating gives $v_N = -u_N < M$ for all such N. This is precisely the definition of $v_N \to -\infty$. Consequently Definition 10.6 gives

$$\liminf s_n = +\infty, \qquad \limsup(-s_n) = -\infty.$$

The convention $-(-\infty) = +\infty$ then proves the desired identity.

Conclusion. In all cases, $\liminf s_n = -\limsup(-s_n)$. The infinite cases follow from tail unboundedness and the infinite-limit definitions, with finite limit laws used only for real tail bounds having a finite limit.

11.9 Exercise 11.9: Closed intervals and an impossible set of subsequential limits

What we must show. (a) Show that $[a, b]$ is closed for real $a \leq b$. (b) Decide whether some real sequence can have exactly $(0, 1)$ as its set of subsequential limits.

Plan. Use the sequential meaning of closed stated with Theorem 11.9: the set contains every limit of a convergent sequence of its points. For part (b), Theorem 11.9 requires the set of subsequential limits itself to have this property.

1. Part (a): Verify the sequential closedness condition

Let (t_k) be a sequence with $t_k \in [a, b]$ for every k, and suppose $t_k \to t \in \mathbb{R}$. Thus

$$a \leq t_k \leq b \qquad (k \in \mathbb{N}).$$

If $t < a$, take $\varepsilon = (a - t)/2 > 0$. Eventually $|t_k - t| < \varepsilon$, so

$$t_k < t + \varepsilon = \frac{a + t}{2} < a,$$

contradicting $t_k \geq a$. If $t > b$, take $\varepsilon = (t-b)/2 > 0$. Eventually $|t_k - t| < \varepsilon$, so

$$t_k > t - \varepsilon = \frac{b+t}{2} > b,$$

contradicting $t_k \leq b$. Both alternatives are impossible. Hence $a \leq t \leq b$, so $t \in [a, b]$. This proves that $[a, b]$ is closed in the required sequential sense.

The argument includes $a = b$: in that case all terms equal a and the only possible limit is a. If extended limits are also considered, neither infinite symbol can arise from such a bounded sequence. For $+\infty$, the positive threshold $\max\{1, b+1\} > b$ cannot be exceeded; for $-\infty$, the negative threshold $\min\{-1, a-1\} < a$ cannot be crossed downwards.

2. Part (b): Apply Theorem 11.9 to a sequence of proposed limits

Suppose, for a contradiction, that a real sequence (s_n) has exactly $S = (0, 1)$ as its set of subsequential limits. Define

$$t_k = \frac{1}{k+1} \qquad (k \in \mathbb{N}).$$

Since $k + 1 \geq 2$, we have $0 < t_k \leq 1/2 < 1$. Thus $t_k \in S$ for every k. Also $t_k \to 0$: for any $\varepsilon > 0$, choose an integer $K > 1/\varepsilon$; then $k > K$ gives

$$|t_k - 0| = \frac{1}{k+1} < \frac{1}{k} < \varepsilon.$$

Theorem 11.9 applies to this sequence of finite members of S and forces its limit 0 to belong to S. But $0 \notin (0, 1)$, a contradiction.

Conclusion. (a) Every closed interval $[a, b]$, including a one-point interval, is closed in the sequential sense. (b) No sequence has $(0, 1)$ as exactly its set of subsequential limits.

11.10 Exercise 11.10: The alternating diagonal traversal in Figure 11.2

What we must show. For the sequence listed by the arrows of Figure 11.2, find (a) its exact set of subsequential limits and (b) its limit superior and limit inferior.

Plan. The figure has the same row repeated infinitely many times. Each reciprocal value therefore occurs at arbitrarily large indices. The top row also supplies a subsequence of values tending to 0. To prove completeness of the answer, isolate any other positive proposed limit from all reciprocal values.

1. Describe the array and its order precisely

Number rows and columns by $i, j \in \mathbb{N}$. The entry in position (i, j) is $1/j$, so the array is

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

The traversal groups the entries by diagonals $i + j = m + 1$, with $m = 1, 2, 3, \ldots$. Diagonal m has exactly m entries. For odd m, the traversal lists columns $1, 2, \ldots, m$; for even m, it lists columns

$m, m-1, \ldots, 1$. Thus the successive diagonal blocks begin

$$\underbrace{1}_{m=1}, \quad \underbrace{\tfrac{1}{2}, 1}_{m=2}, \quad \underbrace{1, \tfrac{1}{2}, \tfrac{1}{3}}_{m=3}, \quad \underbrace{\tfrac{1}{4}, \tfrac{1}{3}, \tfrac{1}{2}, 1}_{m=4}, \quad \ldots.$$

This order agrees with the alternating directions of the arrows.

2. Check explicitly that choosing later diagonals gives later indices

Before diagonal m there are

$$T_{m-1} = 1 + 2 + \cdots + (m-1) = \frac{m(m-1)}{2}$$

listed entries, with $T_0 = 0$. For $m \geq j$, the occurrence of $1/j$ in diagonal m has index

$$I(m, j) = \begin{cases} \dfrac{m(m-1)}{2} + j, & m \text{ odd}, \\ \dfrac{m(m-1)}{2} + (m - j + 1), & m \text{ even}. \end{cases}$$

Indeed, its position within that diagonal is j when columns are in increasing order, and $m - j + 1$ when they are in decreasing order. In both cases

$$T_{m-1} + 1 \leq I(m, j) \leq T_m, \qquad T_m = \frac{m(m+1)}{2}.$$

Every index of diagonal $m+1$ is at least $T_m + 1$, which is strictly larger than every index of diagonal m. Consequently selections made from successive diagonals always have strictly increasing indices.

3. Part (a): Obtain every reciprocal as a subsequential limit

Fix $j \in \mathbb{N}$. Every diagonal $m \geq j$ contains column j once, at the matrix position $(m + 1 - j, j)$. Choose the indices

$$n_k = I(j + k - 1, j) \qquad (k \in \mathbb{N}).$$

They are strictly increasing by Step 2, and the corresponding sequence values satisfy $s_{n_k} = 1/j$ for every k. This constant subsequence converges to $1/j$. Thus every member of $\{1/j : j \in \mathbb{N}\}$ is a subsequential limit.

4. Part (a): Obtain zero from the top row

The top-row entry in column k lies in diagonal k and has value $1/k$. Select it at index

$$n_k = I(k, k) = \begin{cases} \dfrac{k(k+1)}{2}, & k \text{ odd}, \\ \dfrac{k(k-1)}{2} + 1, & k \text{ even}. \end{cases}$$

These indices are strictly increasing because they come from successive diagonals. For example, they begin $1, 2, 6, 7, 15, \ldots$. The selected values are exactly $s_{n_k} = 1/k$. Given $\varepsilon > 0$, an integer $K > 1/\varepsilon$ ensures $k > K$ implies $|1/k - 0| < \varepsilon$. Therefore $s_{n_k} \to 0$, and 0 is also a subsequential limit.

5. Part (a): Exclude every other proposed limit

Every term of the full sequence belongs to $D = \{1/j : j \in \mathbb{N}\} \subseteq (0, 1]$. Hence any finite subsequential limit is in $[0, 1]$, by the same eventual-bound argument used in Exercise 11.5. The bounds also rule out both infinite subsequential limits.

Suppose $t > 0$ and $t \notin D$. We show that a fixed positive neighborhood of t contains no term of the sequence. Choose an integer $J > \max\{2, 2/t\}$. For every $j \geq J$, we have

$$\frac{1}{j} \leq \frac{1}{J} < \frac{t}{2}, \qquad \left|\frac{1}{j} - t\right| = t - \frac{1}{j} > \frac{t}{2}.$$

For the finitely many indices $1 \leq j < J$, none of the values $1/j$ equals t. Their distances from t are strictly positive, so the finite minimum

$$d = \min_{1 \leq j < J} \left|\frac{1}{j} - t\right|$$

exists and is positive. Define $\delta = \min\{t/4, d/2\} > 0$. If $j \geq J$, then $|1/j - t| > t/2 > \delta$. If $j < J$, then $|1/j - t| \geq d > \delta$. Thus every term of the original sequence satisfies $|s_n - t| > \delta$. No subsequence can converge to t, since convergence would require its terms eventually to have distance less than δ from t.

Every possible finite limit is either 0, a member of D, or a positive number not in D. Steps 3–5 account for all these possibilities, and the infinite cases were excluded. Therefore

$$S = \{0\} \cup \left\{\frac{1}{j} : j \in \mathbb{N}\right\}$$

is exactly the set of subsequential limits.

6. Part (b): Use the extreme members of the computed set

The set S has minimum 0 and maximum 1. Theorem 11.8(ii) therefore gives

$$\limsup_{n\to\infty} s_n = \sup S = 1, \qquad \liminf_{n\to\infty} s_n = \inf S = 0.$$

Conclusion. (a) The exact set is $S = \{0\} \cup \{1/j : j \in \mathbb{N}\}$. (b) The limit superior is 1 and the limit inferior is 0.

11.11 Exercise 11.11: An increasing sequence approaching a supremum from the set

What we must show. For a nonempty bounded set $S \subseteq \mathbb{R}$, construct an increasing sequence (s_n) with every $s_n \in S$ and $s_n \to \sup S$.

Plan. If the supremum belongs to the set, use the constant sequence at that point. Otherwise, at each stage choose a point of S above both the previous value and a closer lower approximation to the supremum.

1. State the necessary nonemptiness condition and the meaning of increasing

The exercise's statement needs the implicit assumption $S \neq \varnothing$. For the empty set there is no sequence of points in S, and its supremum is not defined by the real-supremum conventions being used. We therefore assume that S is nonempty and bounded. In fact, the construction needs only that it be bounded above. Completeness gives the finite real number

$$M = \sup S.$$

As in Section 10 and in the note to this exercise, increasing means nondecreasing: $s_n \le s_{n+1}$. A constant sequence is allowed. When $M \notin S$, the construction below will give strict increase.

2. The case in which the supremum belongs to the set

If $M \in S$, define $s_n = M$ for every $n \in \mathbb{N}$. Then all terms belong to S and $s_n = s_{n+1}$, so the sequence is increasing in the stated sense. For every $\varepsilon > 0$,

$$|s_n - M| = 0 < \varepsilon$$

at every index. Hence $s_n \to M$.

3. Begin the construction when M is not a member

Suppose $M \notin S$. Since M is an upper bound, every $s \in S$ satisfies $s \le M$; the nonmembership of M strengthens this to $s < M$ for every $s \in S$. The number $M - 1 < M$ cannot be an upper bound for S, because M is its least upper bound. Therefore some $s_1 \in S$ satisfies

$$M - 1 < s_1 < M.$$

4. Continue the construction with an explicitly admissible threshold

Suppose $s_{n-1} \in S$ has been chosen, where $n \ge 2$. Both $s_{n-1} < M$ and $M - 1/n < M$ hold. Therefore

$$c_n = \max\left\{s_{n-1}, M - \frac{1}{n}\right\} < M.$$

As $c_n < M = \sup S$, the number c_n is not an upper bound for S. Choose $s_n \in S$ with $s_n > c_n$. Every point of S is less than M, so this choice satisfies

$$s_{n-1} < s_n < M, \qquad M - \frac{1}{n} < s_n < M.$$

The procedure can be repeated at every positive integer stage. It gives a sequence in S that is strictly increasing. The displayed approximation bound also holds at $n = 1$ by Step 3.

5. Verify convergence to the supremum

For every n, the construction gives

$$0 < M - s_n < \frac{1}{n}.$$

Let $\varepsilon > 0$. Choose an integer $N > 1/\varepsilon$ by the Archimedean property. If $n > N$, then

$$|s_n - M| = M - s_n < \frac{1}{n} < \varepsilon.$$

Thus $s_n \to M$ by the definition of convergence.

Conclusion. With the necessary hypothesis $S \neq \varnothing$, an increasing sequence in S converging to $\sup S$ always exists. It may be taken constant when $\sup S \in S$, and strictly increasing when $\sup S \notin S$.

12 Limit superior and limit inferior

For each integer $N \geq 0$, the tail after N is the nonempty set

$$E_N = \{s_n : n > N\}.$$

Definition 10.6 gives

$$\limsup s_n = \lim_{N\to\infty} \sup E_N, \qquad \liminf s_n = \lim_{N\to\infty} \inf E_N.$$

These limits may be infinite. When a tail bound is infinite, we check the case through the definitions and extended order before applying any theorem about sequences of real numbers.

A tail supremum need not be attained by a term. Likewise, a term may lie above the lim sup or below the lim inf at arbitrarily large indices. The solutions check the relevant bound and eventuality conditions explicitly.

Tail notation and the meaning of infinite bounds

For a real sequence (x_n) and an integer $N \geq 1$, write

$$X_N = \{x_n : n > N\}, \qquad u_N(x) = \inf X_N, \qquad v_N(x) = \sup X_N.$$

The set X_N is nonempty because it contains x_{N+1}. We use the extended-bound conventions of Section 5: $u_N(x) = -\infty$ if the tail is unbounded below, and $v_N(x) = +\infty$ if it is unbounded above. For a nonempty set of real terms, a tail infimum cannot be $+\infty$, and a tail supremum cannot be $-\infty$. By Definition 10.6,

$$\liminf_{n\to\infty} x_n = \lim_{N\to\infty} u_N(x), \qquad \limsup_{n\to\infty} x_n = \lim_{N\to\infty} v_N(x).$$

Removing more terms enlarges the infimum and decreases the supremum, since $X_{N+1} \subseteq X_N$.

We will also use a finite-prefix observation. If a tail of a real sequence has an upper bound U, then adjoining its finitely many omitted terms still gives a bounded-above set: take the maximum of U and those finitely many terms. The corresponding minimum proves the analogous statement for lower bounds. Thus a sequence that is unbounded above, or unbounded below, has the same property on every tail. Infinite symbols will be handled through order and these conventions, rather than through real-number algebra.

12.1 Exercise 12.1: Eventual comparison passes to both limiting bounds

What we must show. Suppose there is an N_0 such that $s_n \leq t_n$ for all $n > N_0$. Prove

$$\liminf s_n \leq \liminf t_n, \qquad \limsup s_n \leq \limsup t_n.$$

No boundedness assumption is imposed on the two sequences.

Step 1: make the eventual comparison valid on the tails under discussion.

Enlarge N_0, if necessary, to a positive integer. This keeps the hypothesis true, because it discards additional initial terms. Fix any integer $N \geq N_0$. For every $n > N$ we then have $s_n \leq t_n$.

For the upper tail bounds, if $v_N(t) = +\infty$, the extended-order comparison $v_N(s) \le v_N(t)$ holds immediately. Otherwise $v_N(t)$ is real, and

$$s_n \le t_n \le v_N(t) \qquad (n > N).$$

Thus $v_N(t)$ is an upper bound of the tail of (s_n), so its least upper bound satisfies $v_N(s) \le v_N(t)$.

For the lower tail bounds, if $u_N(s) = -\infty$, the comparison $u_N(s) \le u_N(t)$ is automatic in the extended order. Otherwise $u_N(s)$ is real, and

$$u_N(s) \le s_n \le t_n \qquad (n > N).$$

Thus $u_N(s)$ is a lower bound of the tail of (t_n), so it is at most that tail's greatest lower bound. We have proved

$$u_N(s) \le u_N(t), \qquad v_N(s) \le v_N(t) \qquad (N \ge N_0).$$

Step 2: pass to the lower limiting bounds, checking infinite cases.

If (s_n) is unbounded below, every tail infimum $u_N(s)$ equals $-\infty$. Therefore $\liminf s_n = -\infty \le \liminf t_n$, and there is nothing further to prove for the lower bounds in this case.

If (s_n) is bounded below, choose a real lower bound a. For $n > N_0$, the hypothesis gives $t_n \ge s_n \ge a$. The finite-prefix observation shows that (t_n) is bounded below too. Consequently both $u_N(s)$ and $u_N(t)$ are real for every N. Their limits exist in the extended sense by Definition 10.6. Apply Exercise 9.9(c), the order rule for existing limits of real-valued sequences, to $u_N(s) \le u_N(t)$:

$$\lim_{N\to\infty} u_N(s) \le \lim_{N\to\infty} u_N(t).$$

The limits may be $+\infty$, but the terms to which the exercise was applied are real. By definition this is $\liminf s_n \le \liminf t_n$.

Step 3: pass to the upper limiting bounds, checking infinite cases.

If (t_n) is unbounded above, then every $v_N(t) = +\infty$, so $\limsup s_n \le +\infty = \limsup t_n$. If (t_n) is bounded above by a real number b, then $s_n \le t_n \le b$ for $n > N_0$. The finite-prefix observation shows that (s_n) is bounded above as well. Hence $v_N(s)$ and $v_N(t)$ are real-valued sequences, with existing extended limits. Applying Exercise 9.9(c) to their eventual inequality gives

$$\lim_{N\to\infty} v_N(s) \le \lim_{N\to\infty} v_N(t),$$

which is $\limsup s_n \le \limsup t_n$.

Conclusion. Both comparisons hold:

$$\boxed{\liminf s_n \le \liminf t_n, \qquad \limsup s_n \le \limsup t_n}.$$

The proof includes infinite limiting bounds. It does not treat $+\infty$ or $-\infty$ as individual real terms in an application of a real-sequence limit rule.

12.2 Exercise 12.2: A zero limit superior of absolute values

What we must show. Prove the equivalence

$$\limsup |s_n| = 0 \quad \Longleftrightarrow \quad s_n \longrightarrow 0.$$

Forward direction, Step 1: derive the boundedness needed for finite tail suprema.

Assume $\limsup |s_n| = 0$ and put $a_n = |s_n| \geq 0$. If (a_n) were unbounded above, every tail would be unbounded above by the finite-prefix observation. Its every tail supremum would therefore be $+\infty$, giving $\limsup a_n = +\infty$, contrary to the assumed value 0. Thus (a_n) is bounded above. Because all $a_n \geq 0$, it is bounded below as well.

It follows that the numbers

$$V_N = \sup\{|s_n| : n > N\}$$

are finite and nonnegative for every positive integer N. Their nonnegativity follows, for example, by comparing V_N with the member $|s_{N+1}| \geq 0$ of the tail. The hypothesis and Definition 10.6 now say $V_N \to 0$.

Forward direction, Step 2: control every term in one sufficiently late tail.

Let $\varepsilon > 0$. From $V_N \to 0$, choose a positive integer N_* large enough that $|V_{N_*} - 0| < \varepsilon$. Since $V_{N_*} \geq 0$, this means $V_{N_*} < \varepsilon$. For every integer $n > N_*$, the definition of tail supremum gives

$$|s_n - 0| = |s_n| \leq V_{N_*} < \varepsilon.$$

This is Definition 7.1 for $s_n \to 0$. Boundedness was obtained from the hypothesis before any finite-tail estimate was used.

Reverse direction, Step 1: verify convergence of the absolute values.

Assume $s_n \to 0$. Let $\varepsilon > 0$. The definition gives a threshold N such that $n > N$ implies $|s_n - 0| < \varepsilon$. Since $|s_n| \geq 0$, the error in the sequence of absolute values is exactly

$$||s_n| - 0| = |s_n| = |s_n - 0| < \varepsilon.$$

Thus $|s_n| \to 0$ directly from the definition.

Reverse direction, Step 2: identify its limit superior.

Theorem 10.7 says that when a sequence has a limit, its limit inferior and limit superior equal that limit. Applying it to the convergent sequence $(|s_n|)$ gives $\limsup |s_n| = 0$.

Conclusion. The two implications prove $\boxed{\limsup |s_n| = 0 \iff s_n \to 0}$.

12.3 Exercise 12.3: Two synchronized sequences of period four

What we must show. Compute all seven requested combinations for the sequences whose repeated blocks are $(0, 1, 2, 1)$ and $(2, 1, 1, 0)$.

Step 1: keep the phases aligned in one residue table.

The index n has one of four residues modulo 4. Reading the entries at the same index gives

Residue of n	s_n	t_n	$s_n + t_n$	$s_n t_n$
1	0	2	2	0
2	1	1	2	1
3	2	1	3	2
0	1	0	1	0

For instance, at indices $n = 4j + 3$ the entries are $s_n = 2$ and $t_n = 1$, so the sum is 3 and the product is 2. Every row occurs at arbitrarily large indices: by taking the integer j sufficiently large, its index exceeds any prescribed threshold. For the residue 0, use indices $4, 8, 12, \ldots$.

Step 2: identify each tail's minimum and maximum.

Every tail of (s_n) contains the values $0, 1, 2$ and no others. Thus its infimum is 0 and its supremum is 2. The same is true for every tail of (t_n). The sum has the repeated block $(2, 2, 3, 1)$, so every tail of the sum contains exactly the values $1, 2, 3$; its infimum is 1 and its supremum is 3. The product has the repeated block $(0, 1, 2, 0)$, so its tail infima are 0 and its tail suprema are 2. Taking the limits of these constant sequences of tail bounds gives

$$\liminf s_n = 0, \quad \limsup s_n = 2, \qquad \liminf t_n = 0, \quad \limsup t_n = 2,$$

$$\liminf(s_n + t_n) = 1, \quad \limsup(s_n + t_n) = 3, \qquad \liminf(s_n t_n) = 0, \quad \limsup(s_n t_n) = 2.$$

Step 3: evaluate every requested expression.

(a) $\liminf s_n + \liminf t_n = 0 + 0 = \boxed{0}$,

(b) $\liminf(s_n + t_n) = \boxed{1}$,

(c) $\liminf s_n + \limsup t_n = 0 + 2 = \boxed{2}$,

(d) $\limsup(s_n + t_n) = \boxed{3}$,

(e) $\limsup s_n + \limsup t_n = 2 + 2 = \boxed{4}$,

(f) $\liminf(s_n t_n) = \boxed{0}$,

(g) $\limsup(s_n t_n) = \boxed{2}$.

Conclusion. The phase alignment matters. The largest value of (s_n) occurs when $t_n = 1$, whereas the largest value of (t_n) occurs when $s_n = 0$. Adding the separate maxima therefore gives 4, even though the paired sum has maximum 3.

12.4 Exercise 12.4: The limit superior of a sum is at most the sum of the limit superiors

What we must show. For bounded real sequences (s_n) and (t_n), prove

$$\limsup(s_n + t_n) \le \limsup s_n + \limsup t_n.$$

Step 1: establish that all relevant tail bounds are finite.

Choose real constants $A, B \ge 0$ with $|s_n| \le A$ and $|t_n| \le B$ for every n. The triangle inequality gives $|s_n + t_n| \le A + B$, so the sum sequence is bounded as well. For every positive integer N, the three tails are nonempty and bounded, and hence the numbers

$$S_N = \sup\{s_n : n > N\}, \quad T_N = \sup\{t_n : n > N\}, \quad W_N = \sup\{s_n + t_n : n > N\}$$

are real. Their limits as $N \to \infty$ are also finite, since the underlying sequences are bounded.

Step 2: prove the finite-tail inequality.

For every integer $n > N$, the upper-bound properties give $s_n \le S_N$ and $t_n \le T_N$. Adding yields

$$s_n + t_n \le S_N + T_N.$$

Thus the real number $S_N + T_N$ is an upper bound of the paired-sum tail. Its least upper bound satisfies

$$W_N \le S_N + T_N.$$

This argument does not require S_N or T_N to be attained. It also does not require a single index where both separate bounds are approached at once.

Step 3: take limits of the real tail-bound sequences.

By Definition 10.6,

$$S_N \to \limsup s_n, \qquad T_N \to \limsup t_n, \qquad W_N \to \limsup(s_n + t_n).$$

The sum law from Section 9 gives

$$S_N + T_N \longrightarrow \limsup s_n + \limsup t_n.$$

Apply Exercise 9.9(c) to the termwise comparison $W_N \le S_N + T_N$. The resulting comparison of limits is

$$\limsup(s_n + t_n) \le \limsup s_n + \limsup t_n.$$

Conclusion. This proves the desired inequality. The boundedness hypotheses ensure that every sum of limiting bounds used here is a sum of real numbers; no undefined combination of infinite symbols has been introduced.

12.5 Exercise 12.5: The corresponding lower bound for the limit inferior of a sum

What we must show. Use Exercises 11.8 and 12.4 to prove, for bounded real sequences,

$$\liminf(s_n + t_n) \ge \liminf s_n + \liminf t_n.$$

Step 1: verify that the reflected sequences meet the hypotheses.

The sequences $(-s_n)$ and $(-t_n)$ are bounded because $|-s_n| = |s_n|$ and $|-t_n| = |t_n|$. Their sum is $-s_n - t_n = -(s_n + t_n)$, which is bounded too. Exercise 12.4 therefore applies and gives

$$\limsup\big(-(s_n + t_n)\big) \le \limsup(-s_n) + \limsup(-t_n).$$

Step 2: replace each reflected upper limiting bound.

Exercise 11.8 states $\liminf x_n = -\limsup(-x_n)$. For the bounded sequences here, all these values are real. Taking their negatives also gives $\limsup(-x_n) = -\liminf x_n$. Substitute this identity in the three places in Step 1:

$$-\liminf(s_n + t_n) \le -\liminf s_n - \liminf t_n = -\big(\liminf s_n + \liminf t_n\big).$$

Step 3: reverse the inequality by negation.

Multiplying a real inequality by -1 reverses its direction. Negating both sides of the last comparison, and taking the negative twice in each expression, yields

$$\liminf(s_n + t_n) \geq \liminf s_n + \liminf t_n.$$

Conclusion. The claimed lower inequality follows. The sign reversal is the reason that reflecting the upper inequality from Exercise 12.4 produces a lower inequality here.

12.6 Exercise 12.6: Scaling the upper and lower limiting bounds

What we must show. For bounded (s_n) and $k \geq 0$, prove (a) $\limsup(ks_n) = k \limsup s_n$ and (b) $\liminf(ks_n) = k \liminf s_n$. Determine the correct formulas in (c) when $k < 0$.

Preliminary step: scaling preserves boundedness.

Choose $A \geq 0$ with $|s_n| \leq A$ for every n. For any fixed real k, the identity $|ks_n| = |k|\,|s_n| \leq |k|\,A$ shows that (ks_n) is bounded. Its tails are nonempty. Hence all tail bounds for (s_n) and (ks_n) below are real numbers. Write

$$m_N = \inf\{s_n : n > N\}, \qquad M_N = \sup\{s_n : n > N\}.$$

The limits of these two sequences are $\liminf s_n$ and $\limsup s_n$, respectively, and are finite by boundedness.

Parts (a) and (b), the case $k = 0$.

If $k = 0$, every scaled term is zero. Every scaled tail is the singleton $\{0\}$, whose infimum and supremum are zero. Therefore both limiting bounds of (ks_n) are zero. The expressions $0 \cdot \liminf s_n$ and $0 \cdot \limsup s_n$ are also zero, because their other factors are finite real numbers. Thus both requested formulas hold in this case.

Part (a), Step 1: prove the upper-tail scaling formula for $k > 0$.

Fix $k > 0$ and an integer $N \geq 1$. Since $s_n \leq M_N$ for all $n > N$, multiplication by k gives $ks_n \leq kM_N$. Thus kM_N is an upper bound of the scaled tail.

To check that it is the least upper bound, let U be any real upper bound of that tail. Then $ks_n \leq U$ for every $n > N$. Division by $k > 0$ gives $s_n \leq U/k$ for all these indices, so U/k is an upper bound of the original tail. The least-upper-bound property gives $M_N \leq U/k$. Multiplying by $k > 0$ gives $kM_N \leq U$. We have proved both defining properties of a supremum, so

$$\sup\{ks_n : n > N\} = kM_N.$$

Part (a), Step 2: take the limit of the real tail bounds.

By the constant-multiple limit law from Section 9,

$$kM_N \longrightarrow k \limsup s_n.$$

Definition 10.6 identifies the limit of the left side of the tail identity as $\limsup(ks_n)$. Hence

$$\boxed{\limsup(ks_n) = k \limsup s_n} \qquad (k > 0).$$

Together with the zero case, this proves part (a) for all $k \geq 0$.

Part (b), Step 1: prove the lower-tail scaling formula for $k > 0$.

For every $n > N$, $m_N \leq s_n$; multiplying by $k > 0$ gives $km_N \leq ks_n$. Thus km_N is a lower bound of the scaled tail. If L is any real lower bound of that tail, then $L \leq ks_n$ for every $n > N$. Divide by $k > 0$ to obtain $L/k \leq s_n$. Consequently L/k is a lower bound of the original tail. Since m_N is its greatest lower bound, $L/k \leq m_N$, and hence $L \leq km_N$. These two properties establish

$$\inf\{ks_n : n > N\} = km_N.$$

Taking limits of this equality and using the constant-multiple law gives

$$\boxed{\liminf(ks_n) = k \liminf s_n} \qquad (k > 0).$$

The zero case already proved extends it to all $k \geq 0$.

Part (b), Step 2: relate the formula to the reflection hint.

For completeness, the same lower formula follows from part (a) and Exercise 11.8. When $k > 0$,

$$\begin{aligned} \liminf(ks_n) &= -\limsup(-ks_n), \\ &= -\limsup(k(-s_n)), \\ &= -k\limsup(-s_n), \\ &= k\liminf s_n. \end{aligned}$$

The third equality uses part (a) for the bounded sequence $(-s_n)$; the first and last use reflection. All negatives here are negatives of finite real numbers.

Part (c), Step 1: a negative multiplier exchanges the tail bounds.

Now let $k < 0$ and fix N. Multiplication by k reverses order. Since $m_N \leq s_n$ for every $n > N$, we have $ks_n \leq km_N$; thus km_N is an upper bound of the scaled tail. For any other upper bound U, the inequalities $ks_n \leq U$ become $s_n \geq U/k$ after division by $k < 0$. Hence U/k is a lower bound of the original tail, so $U/k \leq m_N$. Multiplying this last inequality by $k < 0$ reverses it and gives $U \geq km_N$. Therefore

$$\sup\{ks_n : n > N\} = km_N \qquad (k < 0).$$

Similarly, $s_n \leq M_N$ gives $ks_n \geq kM_N$, so kM_N is a lower bound of the scaled tail. If L is any other lower bound, $L \leq ks_n$ gives $L/k \geq s_n$ after division by $k < 0$. Thus L/k is an upper bound of the original tail and $M_N \leq L/k$. Multiplying by $k < 0$ gives $kM_N \geq L$. It follows that

$$\inf\{ks_n : n > N\} = kM_N \qquad (k < 0).$$

Part (c), Step 2: take finite limits.

The constant-multiple law, now with a fixed negative constant, still applies to these real convergent tail-bound sequences. Taking limits of the two identities gives

$$\boxed{\limsup(ks_n) = k\liminf s_n, \qquad \liminf(ks_n) = k\limsup s_n} \qquad (k < 0).$$

Conclusion. A positive multiplier preserves the upper and lower limiting bounds separately. A zero multiplier makes both zero. A negative multiplier exchanges their roles, as the two formulas in part (c) state. All statements were proved without assuming that any tail supremum or infimum is attained.

12.7 Exercise 12.7: A positive multiplier preserves an infinite limit superior

What we must show. Suppose $\limsup s_n = +\infty$ and $k > 0$. Prove $\limsup(ks_n) = +\infty$ without assuming that (s_n) is bounded.

Step 1: prove that every tail of the original sequence is unbounded above.

Suppose some tail were bounded above by a real U. Then $s_n \le U$ for every index in that tail, so the eventual comparison result of Exercise 12.1, applied to (s_n) and the constant sequence U, would give

$$\limsup s_n \le \limsup U = U.$$

This contradicts $\limsup s_n = +\infty$. Thus no such tail bound exists: for each integer N and each real R, some index $n > N$ satisfies $s_n > R$.

Step 2: prove that every scaled tail is unbounded above.

Fix any positive integer N and any proposed real upper bound M for the scaled tail. Since $k > 0$, the quotient M/k is real. Step 1, with $R = M/k$, gives an index $n > N$ such that $s_n > M/k$. Multiplying by the positive number k preserves strict inequality:

$$ks_n > M.$$

Thus M is not an upper bound. Because M was arbitrary, the scaled tail is unbounded above. Because N was arbitrary, every scaled tail has this property.

Step 3: apply the definition of the infinite tail bound.

For every N, the extended-bound convention gives

$$\sup\{ks_n : n > N\} = +\infty.$$

Its limiting value is therefore $+\infty$, which is exactly $\limsup(ks_n)$ by Definition 10.6.

Conclusion. We obtain $\boxed{\limsup(ks_n) = +\infty}$. The proof divided and multiplied only real numbers, such as M and M/k, and used unboundedness to interpret the infinite result.

12.8 Exercise 12.8: A product bound for bounded nonnegative sequences

What we must show. Let (s_n) and (t_n) be bounded sequences with $s_n \ge 0$ and $t_n \ge 0$ for every n. Prove

$$\limsup(s_n t_n) \le (\limsup s_n)(\limsup t_n).$$

Step 1: give finite bounds for the three sequences.

Choose $A, B \ge 0$ such that $0 \le s_n \le A$ and $0 \le t_n \le B$ for every n. Because $t_n \ge 0$, multiplication of $s_n \le A$ by t_n gives $s_n t_n \le A t_n$. Because $A \ge 0$, multiplication of $t_n \le B$ by A gives $A t_n \le AB$. Thus

$$0 \le s_n t_n \le A t_n \le AB.$$

The product sequence is bounded and nonnegative too. For every N, all three tails are nonempty and bounded, and their suprema

$$S_N = \sup\{s_n : n > N\}, \quad T_N = \sup\{t_n : n > N\}, \quad P_N = \sup\{s_n t_n : n > N\}$$

are finite and nonnegative. For example, $0 \le s_{N+1} \le S_N$, which proves $S_N \ge 0$; the other two nonnegativity statements follow in the same way from their tails.

Step 2: prove the paired-product tail inequality.

For each $n > N$, the tail upper bounds give $s_n \le S_N$ and $t_n \le T_N$. Multiply the first inequality by $t_n \ge 0$:

$$s_n t_n \le S_N t_n.$$

Multiply the second by $S_N \ge 0$:

$$S_N t_n \le S_N T_N.$$

The sign checks are necessary for both comparisons. Combining them shows $s_n t_n \le S_N T_N$ for every $n > N$. Hence $S_N T_N$ is an upper bound of the product tail. Its least upper bound satisfies

$$P_N \le S_N T_N.$$

Neither separate tail supremum needs to be attained, and no common index attaining both was selected.

Step 3: compare the finite limits of the tail-bound sequences.

Definition 10.6 gives

$$S_N \to \limsup s_n, \qquad T_N \to \limsup t_n, \qquad P_N \to \limsup(s_n t_n).$$

All limits here are finite because the original sequences and the product sequence are bounded. The product limit law from Section 9 therefore yields

$$S_N T_N \longrightarrow (\limsup s_n)(\limsup t_n).$$

Applying Exercise 9.9(c) to $P_N \le S_N T_N$ gives the requested comparison of limiting values.

Conclusion. We have proved

$$\boxed{\limsup(s_n t_n) \le (\limsup s_n)(\limsup t_n)}.$$

The proof includes the case where either upper limiting bound is zero, because no division by one of those bounds was used.

12.9 Exercise 12.9: A positive lower bound and products tending upward

What we must show. Prove the product assertions in parts (a) and (b), and identify Exercise 12.7 as the special case requested in part (c).

A preliminary fact: extracting one positive lower bound.

Write $\alpha = \liminf t_n$, and suppose $\alpha > 0$. The value $\alpha = +\infty$ is allowed. We claim that there are a real number $c > 0$ and an integer N_0 such that

$$t_n \ge c \qquad (n > N_0).$$

If α is finite, choose $c = \alpha/2$; then $0 < c < \alpha$. If $\alpha = +\infty$, choose $c = 1$ instead. In either case, the tail infima $\inf\{t_n : n > N\}$ eventually exceed c, by their convergence to α. Each term in such a tail is

at least its infimum, and therefore exceeds c. Replacing this strict inequality by $\geq c$ gives the claim. This argument also covers an initially unbounded sequence: since $\liminf t_n > 0$, some tail has a finite lower bound.

(a) An ordinary limit of $+\infty$.

Assume $s_n \to +\infty$. To prove $s_n t_n \to +\infty$, fix an arbitrary real threshold $M > 0$. By the definition of this limit, there is N_1 such that

$$n > N_1 \quad \Longrightarrow \quad s_n > M/c.$$

Since $M/c > 0$, these s_n are positive. If $n > \max\{N_0, N_1\}$, we can multiply $t_n \geq c$ by the positive number s_n, obtaining

$$s_n t_n \geq c s_n > c(M/c) = M.$$

Thus every sufficiently late product exceeds M. Checking positive thresholds suffices: exceeding any positive number also exceeds all smaller, possibly negative, thresholds. Hence $s_n t_n \to +\infty$.

(b) A limit superior of $+\infty$.

Now assume only $\limsup s_n = +\infty$. Every tail of (s_n) is unbounded above. Indeed, a tail bounded above, together with the finitely many preceding real terms, would bound the whole sequence above and force its limit superior to be at most a real upper bound.

Fix a tail index N and a positive threshold M. The unbounded tail beyond $\max\{N, N_0\}$ contains an index n for which $s_n > M/c$. At that index $n > N$, and the same sign calculation as in part (a) gives $s_n t_n > M$. Thus every tail of the product sequence is unbounded above. Its tail suprema are all $+\infty$, so its limit superior is $+\infty$.

The distinction between (a) and (b) is in the quantifiers: part (a) puts every sufficiently late product above M; part (b) produces at least one such product in every tail.

(c) Recover Exercise 12.7.

If $t_n = k > 0$ for every n, then $\liminf t_n = k > 0$ and $s_n t_n = k s_n$. Part (b) therefore states exactly that

$$\limsup s_n = +\infty \quad \Longrightarrow \quad \limsup(k s_n) = +\infty \qquad (k > 0).$$

Conclusion. Both product conclusions hold; a positive constant second factor gives Exercise 12.7.

12.10 Exercise 12.10: Detecting boundedness with the limit superior of absolute values

What we must show. Prove both directions of

$$(s_n) \text{ is bounded} \quad \Longleftrightarrow \quad \limsup |s_n| < +\infty.$$

Plan. Absolute values express upper and lower bounds on s_n in one nonnegative sequence. An eventual bound becomes a bound on all terms after we include the finite initial segment.

A bounded sequence gives a finite limit superior.

If (s_n) is bounded, there is a finite $B \geq 0$ with $|s_n| \leq B$ for every n. Thus for every N,

$$0 \leq \sup\{|s_n| : n > N\} \leq B.$$

These tail suprema decrease as N increases. The monotone sequence theorem gives a limit in $[0, B]$, and by definition that limit is $\limsup |s_n|$. In particular it is less than $+\infty$.

A finite limit superior gives an eventual bound.

Conversely, suppose $L = \limsup |s_n| < +\infty$. Since all the absolute values are nonnegative, $L \geq 0$, so L is a real number. The sequence $(|s_n|)$ cannot be unbounded above: if it were, every tail would be unbounded above, since deleting finitely many real terms cannot remove unboundedness. All its tail suprema would then be $+\infty$, contradicting the finite value of L.

Equivalently, some tail has a finite supremum. Once that happens, all subsequent tail suprema are finite and converge to L. Their convergence yields an integer $N \geq 1$ with

$$|s_n| \leq \sup\{|s_j| : j > N\} < L + 1 \qquad (n > N).$$

Include the early terms.

The finite set of real numbers $\{|s_1|, \ldots, |s_N|, L+1\}$ has a finite maximum B. If $n \leq N$, the definition of B gives $|s_n| \leq B$. If $n > N$, the tail estimate gives $|s_n| < L + 1 \leq B$. Consequently $|s_n| \leq B$ for every index n.

Conclusion. The two conditions are equivalent. The limit superior bounds the tail, and a finite maximum absorbs the initial terms.

12.11 Exercise 12.11: The lower ratio–root inequality

What we must show. For a sequence with $s_n \neq 0$ at every index, prove the first inequality of Theorem 12.2:

$$\liminf \left| \frac{s_{n+1}}{s_n} \right| \;\leq\; \liminf |s_n|^{1/n}.$$

Plan. A lower bound on successive ratios becomes a geometric lower bound on the terms. Taking nth roots removes the fixed factor.

Name the two nonnegative extended limits.

Set

$$A = \liminf \left| \frac{s_{n+1}}{s_n} \right|, \qquad B = \liminf |s_n|^{1/n}.$$

Every ratio is well defined and positive because no s_n is zero. The roots are positive as well. Therefore $A, B \in [0, +\infty]$. If $A = 0$, then $A \leq B$ follows immediately from $B \geq 0$.

An arbitrary positive number below A is an eventual ratio bound.

Suppose $A > 0$, and choose any real a with $0 < a < A$. When $A = +\infty$, this means that a can be any positive real number. The definition of the limit inferior gives an integer $K \geq 1$ such that

$$\left| \frac{s_{j+1}}{s_j} \right| > a \qquad (j \geq K).$$

To see this directly, some tail infimum of these ratios exceeds a; each ratio in that tail is at least the infimum. Increase the starting index to K if needed.

Multiply the consecutive inequalities.

For $n > K$, cancellation of the nonzero terms gives

$$|s_n| = |s_K| \prod_{j=K}^{n-1} \left| \frac{s_{j+1}}{s_j} \right| \geq |s_K|\, a^{n-K}.$$

The weak inequality also holds when $n = K$, since the product then has no factors and equals 1. Define the fixed positive constant $C = |s_K|\, a^{-K}$. The estimate becomes

$$|s_n| \geq Ca^n \qquad (n \geq K).$$

The map $x \mapsto x^{1/n}$ is increasing on nonnegative reals, so

$$|s_n|^{1/n} \geq aC^{1/n} \qquad (n \geq K).$$

Pass to the limit inferior and then let a approach A.

Since $C > 0$, the elementary root limit gives $C^{1/n} \to 1$. Thus $aC^{1/n} \to a$, and eventual order (Exercise 12.1) yields $B \geq a$.

If A is finite and $B < A$, then B is finite and a number a strictly between B and A satisfies $a > 0$. This contradicts $B \geq a$. Hence $B \geq A$ in the finite case. If $A = +\infty$, the inequality $B \geq a$ holds for every positive real a; a finite B would be contradicted by taking $a > B$. Therefore $B = +\infty$.

Conclusion. In every case the lower ratio limit inferior is at most the root limit inferior. This proof does not rely on Corollary 12.3, whose proof uses the inequality being established here.

12.12 Exercise 12.12: Arithmetic means and the limits of the original sequence

What we must show. For $s_n \geq 0$, put

$$\sigma_n = \frac{s_1 + \cdots + s_n}{n}.$$

Prove the four-term inequality in part (a), deduce part (b), and give the counterexample requested in part (c).

(a), preparation: name the bounds and check nonnegativity.

Write $\alpha = \liminf s_n$ and $\beta = \limsup s_n$. Because $s_n \geq 0$, all averages are nonnegative, and $\alpha, \beta, \liminf \sigma_n, \limsup \sigma_n$ belong to $[0, +\infty]$. The inequality $\liminf \sigma_n \leq \limsup \sigma_n$ holds for every sequence by Theorem 10.7. We prove the two outside inequalities separately.

(a), upper bound: verify the estimate in the hint.

First suppose $\beta < +\infty$. The sequence (s_n) must then be bounded above: otherwise every tail would be unbounded above, and its limit superior would be $+\infty$. Set

$$U_N = \sup\{s_n : n > N\}, \qquad P_N = s_1 + \cdots + s_N.$$

These are finite, with $U_N, P_N \geq 0$. If $n > N$, each of the $n - N$ remaining terms is at most U_N, so

$$\sigma_n \leq \frac{P_N}{n} + \frac{n-N}{n} U_N \leq \frac{P_N}{n} + U_N.$$

The second inequality uses $U_N \geq 0$ and $(n-N)/n \leq 1$. If $n > M > N$, the inequality $P_N/n \leq P_N/M$ uses $P_N \geq 0$. Taking the supremum over $n > M$ gives the hint exactly:

$$\sup\{\sigma_n : n > M\} \leq \frac{P_N}{M} + U_N.$$

Hold N fixed and let $M \to \infty$. The left side tends to $\limsup \sigma_n$, while $P_N/M \to 0$, yielding $\limsup \sigma_n \leq U_N$. This holds for every N. Letting $N \to \infty$ and using $U_N \to \beta$, we get $\limsup \sigma_n \leq \beta$. All the relevant quantities are finite here, since a global bound on s_n also bounds each average.

If instead $\beta = +\infty$, the desired upper bound is simply $\limsup \sigma_n \leq +\infty$, which always holds. No finite-tail estimate involving an infinite supremum is required.

(a), lower bound: retain a uniformly positive tail.

If $\alpha = 0$, nonnegativity already gives $\liminf \sigma_n \geq 0 = \alpha$. If $\alpha > 0$, choose an arbitrary real c with $0 < c < \alpha$. For $\alpha = +\infty$, take any $c > 0$. There is N such that $s_j \geq c$ for all $j > N$, by the definition of the limit inferior. For $n > N$, discard the nonnegative initial sum and estimate the tail:

$$\sigma_n = \frac{P_N + s_{N+1} + \cdots + s_n}{n} \geq \frac{(n-N)c}{n} = c\left(1 - \frac{N}{n}\right).$$

The expression on the right tends to c, since N is fixed. Exercise 12.1 gives $\liminf \sigma_n \geq c$. When α is finite, letting c range over all positive numbers below α forces $\liminf \sigma_n \geq \alpha$: otherwise choose c between these two quantities. When $\alpha = +\infty$, the same inequality for every $c > 0$ forces $\liminf \sigma_n = +\infty$. Combining the three inequalities proves

$$\boxed{\liminf s_n \leq \liminf \sigma_n \leq \limsup \sigma_n \leq \limsup s_n.}$$

(b) Convergence passes to the averages.

Suppose $s_n \to L$. Nonnegativity means that either $L \geq 0$ is finite or $L = +\infty$; a limit of $-\infty$ is impossible. Then $\liminf s_n = \limsup s_n = L$, and part (a) gives

$$L \leq \liminf \sigma_n \leq \limsup \sigma_n \leq L.$$

Thus both middle quantities equal L. Theorem 10.7, which characterizes convergence by equality of the limit inferior and limit superior, implies $\sigma_n \to L$. If $L = +\infty$, this statement means that the averages also tend to $+\infty$.

(c) Convergent averages need not imply convergent terms.

Take $s_n = 1 + (-1)^n$. Its terms alternate $0, 2, 0, 2, \ldots$, and are nonnegative. The odd subsequence is constantly 0 and the even subsequence is constantly 2, so (s_n) has no limit.

For $n = 2k$, there are k zeros and k twos, so $\sigma_{2k} = 2k/(2k) = 1$. For $n = 2k - 1$, there are k zeros and $k - 1$ twos, so

$$\sigma_{2k-1} = \frac{2(k-1)}{2k-1} = 1 - \frac{1}{2k-1}.$$

In both cases $|\sigma_n - 1| \leq 1/n$. Given $\varepsilon > 0$, choose $N > 1/\varepsilon$; then $n > N$ implies $|\sigma_n - 1| \leq 1/n < \varepsilon$. Thus $\sigma_n \to 1$.

Conclusion. The averages satisfy the four-term bounds and inherit every limit of a nonnegative sequence. The converse fails for alternating zeros and twos.

12.13 Exercise 12.13: Limit inferior and superior as eventual bounds

What we must show. For a bounded real sequence define

$$A = \{a \in \mathbb{R} : \{n \in \mathbb{N} : s_n < a\} \text{ is finite}\},$$
$$B = \{b \in \mathbb{R} : \{n \in \mathbb{N} : s_n > b\} \text{ is finite}\}.$$

Prove $\sup A = \liminf s_n$ and $\inf B = \limsup s_n$.

Plan. A finite set of exceptional indices can be placed before a single cutoff. Thus A consists of eventual lower bounds and B of eventual upper bounds. The boundary value itself need not belong to the respective set.

Check existence of the real suprema and infima.

Choose $C \geq 0$ with $-C \leq s_n \leq C$ for every n. For $a = -C$, no term satisfies $s_n < a$, so $-C \in A$. If $a > C$, every index satisfies $s_n < a$, so $a \notin A$. Thus A is nonempty and bounded above by C. Similarly, $C \in B$, and every $b < -C$ lies outside B, since $s_n > b$ for every n. Hence B is nonempty and bounded below by $-C$. Completeness gives finite real values $\sup A$ and $\inf B$. Write

$$\alpha = \liminf s_n, \quad \beta = \limsup s_n, \quad L_N = \inf\{s_n : n > N\}, \quad U_N = \sup\{s_n : n > N\}.$$

All these numbers are finite, and $L_N \to \alpha$, $U_N \to \beta$.

Every eventual lower bound is at most α.

If $a \in A$, its finitely many exceptional indices lie at most at some index N_0; if there are none, take $N_0 = 1$. Then $s_n \geq a$ whenever $n > N_0$. For $N \geq N_0$, it follows that $L_N \geq a$, and passage to the limit gives $\alpha \geq a$. Thus α is an upper bound of A, so $\sup A \leq \alpha$.

No number below α can be an upper bound of A.

If $a < \alpha$, convergence $L_N \to \alpha$ implies $L_N > a$ for some N. For every $n > N$, we have $s_n \geq L_N > a$. Consequently the indices with $s_n < a$ are contained in $\{1, \ldots, N\}$, a finite set; hence $a \in A$. Now fix any $d < \alpha$ and take $a = (d + \alpha)/2$. Then $d < a < \alpha$, so $a \in A$ and $a > d$. Therefore d fails to be an upper bound of A. This proves that α is the least upper bound: $\sup A = \alpha$.

Repeat the two directions for eventual upper bounds.

If $b \in B$, there is N_0 such that $s_n \leq b$ whenever $n > N_0$. Then $U_N \leq b$ for all $N \geq N_0$; taking limits gives $\beta \leq b$. Thus β is a lower bound of B.

If $b > \beta$, convergence $U_N \to \beta$ gives some N with $U_N < b$. For every $n > N$, we have $s_n \leq U_N < b$, so the indices satisfying $s_n > b$ form a finite set. Thus $b \in B$. If $d > \beta$, the number $b = (\beta + d)/2$ satisfies $\beta < b < d$ and belongs to B, so d cannot be a lower bound of B. Hence β is its greatest lower bound: $\inf B = \beta$.

Why the boundary is not asserted to lie in the set.

For $s_n = -1/n$, one has $\alpha = 0$, but every $s_n < 0$, so $0 \notin A$. For $s_n = 1/n$, one has $\beta = 0$, but every $s_n > 0$, so $0 \notin B$. Suprema and infima need not be attained.

Conclusion. The least eventual upper bound is $\limsup s_n$, and the greatest eventual lower bound is $\liminf s_n$, in the precise supremum/infimum sense above.

12.14 Exercise 12.14: The growth of the geometric mean of the first integers

What we must show. Calculate the limits of $(n!)^{1/n}$ and $(n!)^{1/n}/n$.

Plan. Apply the ratio–root corollary to two different positive sequences. For the second limit we also justify the elementary sequence that defines e, without using Stirling's formula.

(a) Apply the ratio–root corollary to the factorials.

Let $v_n = n!$, which is positive. Cancelling the common product $1 \cdot 2 \cdots n$ gives

$$\frac{v_{n+1}}{v_n} = \frac{(n+1)!}{n!} = n+1 \longrightarrow +\infty.$$

Corollary 12.3 states that a limit of the absolute successive ratios is also the limit of the nth roots, including the value $+\infty$. Therefore

$$\boxed{\lim_{n\to\infty} (n!)^{1/n} = +\infty.}$$

(b), preparation: justify the sequence defining e.

Set $b_n = (1 + 1/n)^n$. The standard constant e is the limit of this sequence, as described in the text's Section 7, Example 1(e). For completeness we verify that this limit exists and is positive. By the binomial theorem,

$$b_n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} = \sum_{k=0}^{n} \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

For $k = 0$ the product is empty and equals 1. For fixed $k \le n$, each displayed factor is nonnegative and does not decrease when n is replaced by $n+1$. Thus each of the old summands does not decrease. The extra summand with $k = n+1$ in the expression for b_{n+1} is positive, so $b_{n+1} > b_n$.

Every factor is at most 1. For $k \ge 2$, the $k-1$ factors $2, 3, \ldots, k$ in $k!$ are each at least 2, whence $k! \ge 2^{k-1}$. It follows that

$$b_n \le \sum_{k=0}^{n} \frac{1}{k!} \le 2 + \sum_{k=2}^{n} \frac{1}{2^{k-1}} < 3.$$

For $n = 1$ the last sum is empty, and the same bound holds. Since $b_1 = 2$, the monotone sequence theorem now gives $b_n \to e$ for a finite number $e \ge 2 > 0$.

(b), normalization: put the denominator inside the root.

Define $u_n = n!/n^n > 0$. Its positive nth root is

$$u_n^{1/n} = \frac{(n!)^{1/n}}{n},$$

because the nth power of the expression on the right is $n!/n^n$ and the positive nth root is unique. The successive ratio simplifies one factor at a time:

$$\begin{aligned} \frac{u_{n+1}}{u_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} \\ &= \frac{(n+1)n^n}{(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n} = \frac{1}{b_n}. \end{aligned}$$

Since $b_n \to e \neq 0$, the reciprocal limit law gives $u_{n+1}/u_n \to 1/e$. Corollary 12.3 therefore gives $u_n^{1/n} \to 1/e$, which is exactly

$$\boxed{\lim_{n\to\infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}.}$$

Conclusion. The unnormalized geometric mean tends to $+\infty$; after division by n, it tends to $1/e$.